



NSW Education Standards Authority

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Centre Number

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Student Number

2025 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Centre Number and Student Number at the top of this page

Total marks:
100

Section I – 20 marks (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–16)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Which sector of the Australian food industry includes the production of crops and livestock?

 - A. Food retail
 - B. Food manufacturing
 - C. Agriculture and fisheries
 - D. Food service and catering

- 2** What is the first step of the food product development process?

 - A. Idea generation
 - B. Market research
 - C. Product specification
 - D. Prototype development

- 3** What is the function of a humectant in a food product?

 - A. To improve colour
 - B. To increase viscosity
 - C. To reduce moisture loss
 - D. To restore flavour damage

- 4** What is the primary reason for a food manufacturer conducting a product recall?

 - A. To adjust pricing for perishable food items
 - B. To promote new food products to consumers
 - C. To reduce food waste in the production process
 - D. To remove non-compliant food products from the market

- 5 Which of the following nutrients is primarily known for its antioxidant properties?
- A. Iron
 - B. Protein
 - C. Calcium
 - D. Vitamin C
- 6 Which piece of legislation is being breached when an untrained worker is instructed to operate a delicatessen meat slicer?
- A. *Food Act 2003* (NSW)
 - B. *Work Health and Safety Act 2011* (NSW)
 - C. *Trade Practices Act 1974* (Commonwealth)
 - D. *Competition and Consumer Act 2010* (Commonwealth)
- 7 A popular confectionery company has released a range of different flavoured chocolate bars designed by celebrity chefs.
- What type of food product development is this an example of?
- A. Me too
 - B. Copycat
 - C. Line extension
 - D. New to the world
- 8 Which of the following food products has had the most value added to it during food manufacture?
- A. Potato chips
 - B. Mashed potato
 - C. Unwashed potatoes
 - D. Meat and potato pie

- 9** A person visits the doctor with the following symptoms: weakness, pale skin, fatigue and loss of appetite.
- Which of the following dietary-related conditions are they most likely to be suffering from?
- A. Obesity
 - B. Anaemia
 - C. Hypertension
 - D. Coronary heart disease
- 10** Which of the following is responsible for regulating the use of genetically modified (GM) foods in Australia?
- A. Australia Food and Grocery Council (AFGC)
 - B. Food Standards Australia New Zealand (FSANZ)
 - C. Australian Competition and Consumer Commission (ACCC)
 - D. Commonwealth Scientific and Industrial Research Organisation (CSIRO)
- 11** Which of the following statements about food allergy or food intolerance is true?
- A. A food allergy is rarely detectable at birth.
 - B. A food intolerance is always life threatening.
 - C. A food allergy causes an autoimmune response in the body.
 - D. A food intolerance usually causes an anaphylactic response.
- 12** What type of equipment is most likely used for food preparation in a small bakery with five employees?
- A. Manual
 - B. Automated
 - C. Large scale
 - D. Computerised

- 13** Why are low glycaemic index (GI) foods recommended for people with type 2 diabetes?
- A. They increase blood sugar levels quickly to boost energy.
 - B. They contain no carbohydrates and reduce blood sugar levels.
 - C. They assist in maintaining stable blood sugar levels over time.
 - D. They increase the need for insulin in managing blood sugar levels.
- 14** Which piece of commercial equipment is used to separate cream from whole milk?
- A. Centrifuge
 - B. Filtration system
 - C. Extrusion device
 - D. Commercial sieve
- 15** What type of packaging uses an oxygen absorber to modify the environment within the package?
- A. Gas
 - B. Active
 - C. Aseptic
 - D. Barrier specific
- 16** At what stage of the food product life cycle would a product reach market saturation?
- A. Decline
 - B. Growth
 - C. Introduction
 - D. Maturity

- 17** Which of the following is an example of primary market research?
- A. Analysing consumer feedback data from reviews
 - B. Using sales data from similar products
 - C. Conducting an online focus group
 - D. Reviewing government reports
- 18** What type of dryer machinery would NOT be suitable to use in the commercial production of breakfast cereal flakes?
- A. Spray
 - B. Drum
 - C. Tunnel
 - D. Cabinet
- 19** In the production of bread, what gas is produced during yeast fermentation?
- A. Oxygen
 - B. Sulphur
 - C. Nitrogen
 - D. Carbon dioxide
- 20** A new food product is priced lower to attract a larger customer base and increase its market share.

What type of pricing strategy is this an example of?

- A. Bundle pricing
- B. Price skimming
- C. Premium pricing
- D. Penetration pricing

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page
- Use this answer booklet for Sections II, III and IV
- Follow the instructions inside this booklet as to where you should write your answers

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (6 marks)

Answer parts (a) and (b) with reference to a named organisation in the Australian food industry that you have studied.

Name of organisation:

- (a) Outline a research and development activity carried out by the organisation. **2**

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- (b) Describe how lifestyle changes of consumers have been addressed by the organisation. **4**

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Question 22 (9 marks)

- (a) Identify TWO reasons why some food manufacturers prefer to use plastic packaging. 2

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- (b) Describe TWO examples of tamper-evident packaging that can be used by a food manufacturer to ensure the safety of its products in retail facilities. 3

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- (c) Explain the importance of accurate food labelling for consumers. 4

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Question 23 (11 marks)

Answer parts (a) and (b) with reference to a specific group in Australia that has nutritional considerations.

Name of group:

- (a) Describe a strategy used to promote optimum health through good nutrition for this group.

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Question 23 continues on page 13

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Question 23 (continued)

- (b) A breakfast meal plan is shown.

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Breakfast meal plan	
Porridge:	Rolled oats, full cream milk, brown sugar
Toast:	White bread, butter, strawberry jam
Coffee:	Black

Justify TWO modifications to the breakfast meal plan that could address the specific nutritional needs of the named group.

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- (c) Describe how the economic environment can affect the development of new food products in the Australian food industry.

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End of Question 23

Question 24 (7 marks)

- (a) Outline TWO reasons why a can of tomatoes may bulge after it has been processed.

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- (b) Describe quality management procedures in relation to the commercial production of a food product.

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Question 25 (9 marks)

- (a) Explain the benefits of preserving food.

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- (b) Compare the effectiveness of temperature control with ONE other preservation principle.

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Section III — Answer Question 27 on pages 18–24

27

Question Number

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If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 25–31

28

Question Number

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 18–24 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

An ice cream company would like to develop a new flavour using native plant ingredients. The ice cream will be distributed to supermarkets nationwide. The company will need to invest in new equipment.

- | | | |
|-----|---|----------|
| (a) | Describe ONE additive that could be used during the commercial manufacture of this new ice cream. | 2 |
| (b) | Conduct a SWOT analysis of this new product for the manufacturing company. | 5 |
| (c) | Propose a marketing plan that could be used to market this new ice cream. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 25–31 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
-

Question 28 (15 marks)

Explain how a person's lifestyle, culture and social practices can affect their nutritional status.

End of paper